

20262822 姚翔 Linux命令实践

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实践环境：WSL Ubuntu 终端（普通用户 yaoxiang）

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一、学习内容记录

Linux 命令由命令名、选项和参数组成。命令名说明要执行的操作；选项通常以 - 或 -- 开头，用于改变命令行为；参数表示命令处理的文件、目录或其他对象。例如，`ls -l /home` 中，`ls` 为命令名，`-l` 为选项，`/home` 为参数。

本次学习并实践了目录查看与切换命令（`ls`、`pwd`、`cd`），登录与帮助查询命令（`who`、`man`、`whereis`、`which`），文件查找与文本检索命令（`find`、`locate`、`grep`），以及文本和二进制处理命令（`echo`、`od`、`sort`）。另外还了解了文件管理、磁盘管理、进程管理、用户管理、网络管理和压缩解压等常用命令类别。

二、实践过程与结果

1. `ls`：列出目录内容

依次实践了 `ls`、`ls -l`、`ls -a`、`ls -lh`、`ls /root`、`ls -t` 和 `ls -R`。前四项和按时间排序、递归列出目录均成功。`ls /root` 失败，终端提示 `Permission denied`，因为当前使用的是普通用户，默认没有读取 `root` 用户家目录的权限；如确有管理需要，应在确认权限范围后使用 `sudo ls /root`。

```
yaoliang@DESKTOP-HJNS067:~$ ls
afl-2.57b  afl-exp  afl-test  test.txt
yaoliang@DESKTOP-HJNS067:~$ ls -l
total 16
drwxr-xr-x 12 yaoliang yaoliang 4096 Aug  3 15:08 afl-2.57b
drwxr-xr-x  7 yaoliang yaoliang 4096 Aug  4 15:07 afl-exp
drwxr-xr-x  4 yaoliang yaoliang 4096 Aug  3 15:15 afl-test
-rw-r--r--  1 yaoliang yaoliang 13 Sep 21 08:32 test.txt
yaoliang@DESKTOP-HJNS067:~$ ls -a
.  .bash_history  .bashrc  .config  .lessshst  .profile  afl-2.57b  afl-test
.. .bash_logout  .cache  .landscape  .motd_shown  wget-hsts  afl-exp  test.txt
yaoliang@DESKTOP-HJNS067:~$ ls -lh
total 16K
drwxr-xr-x 12 yaoliang yaoliang 4.0K Aug  3 15:08 afl-2.57b
drwxr-xr-x  7 yaoliang yaoliang 4.0K Aug  4 15:07 afl-exp
drwxr-xr-x  4 yaoliang yaoliang 4.0K Aug  3 15:15 afl-test
-rw-r--r--  1 yaoliang yaoliang 13 Sep 21 08:32 test.txt
ls: cannot open directory '/root': Permission denied
yaoliang@DESKTOP-HJNS067:~$ ls -lt
test.txt  afl-exp  afl-test  afl-2.57b
yaoliang@DESKTOP-HJNS067:~$ ls -R
.:
afl-2.57b  afl-exp  afl-test  test.txt

./afl-2.57b:
Android.bp          afl-as             afl-fuzz.c         afl-showmap        as                 libdislocator     types.h
CONTRIBUTING.md   afl-as.c          afl-g++           afl-showmap.c     config.h          libtkencaap
LICENSE            afl-as.h          afl-gcc           afl-tmin           debug.h           llvm_mode
Makefile            afl-clang          afl-gcc.c         afl-tmin.c         dictionaries      qemu_mode
README.md           afl-clang++       afl-gotcpu        afl-whatsup        docs              test-instr.c
afl-analyze        afl-cmin          afl-gotcpu.c      alloc-inl.h       experimental      test-libfuzzer-target.c
afl-analyze.c      afl-fuzz          afl-plot          android-ashmem.h   hash.h            testcases

./afl-2.57b/dictionaries:
README.dictionaries  html_tags.dict  js.dict  pdf.dict  png.dict  sql.dict  webp.dict
gif.dict             jpeg.dict      json.dict  perl.dict  regexp.dict  tiff.dict  xml.dict

./afl-2.57b/docs:
COPYING  QuickStartGuide.txt  life_pro_tips.txt  perf_tips.txt  technical_details.txt
Changelog  env_variables.txt  notes_for_asan.txt  sister_projects.txt  visualization
INSTALL  historical_notes.txt  parallel_fuzzing.txt  status_screen.txt  vuln_samples

./afl-2.57b/docs/visualization:
afl_gzip.png

./afl-2.57b/docs/vuln_samples:
bash-cmd-exec.var          libtiff-bad-write.tif  sqlite-bad-ptr3.sql  sqlite-null-ptr8.sql
bash-uninit-mem.var       libtiff-uninit-mem.tif  sqlite-heap-overflow.sql  sqlite-null-ptr9.sql
ffmpg-h264-bad-read.mp4  libtiff-uninit-mem2.tif  sqlite-heap-overflow.sql  sqlite-oob-read.sql
ffmpg-h264-bad-read.mp4  libtiff-uninit-mem3.tif  sqlite-negative-memset.sql  sqlite-oob-write.sql
ffmpg-h264-call-stack-overflow.mp4  libtiff-uninit-mem4.tif  sqlite-null-ptr1.sql  sqlite-stack-buf-overflow.sql

./afl-2.57b/dictionaries:
README.dictionaries  html_tags.dict  js.dict  pdf.dict  png.dict  sql.dict  webp.dict
gif.dict             jpeg.dict      json.dict  perl.dict  regexp.dict  tiff.dict  xml.dict

./afl-2.57b/docs:
COPYING  QuickStartGuide.txt  life_pro_tips.txt  perf_tips.txt  technical_details.txt
Changelog  env_variables.txt  notes_for_asan.txt  sister_projects.txt  visualization
INSTALL  historical_notes.txt  parallel_fuzzing.txt  status_screen.txt  vuln_samples

./afl-2.57b/docs/visualization:
afl_gzip.png

./afl-2.57b/docs/vuln_samples:
bash-cmd-exec.var          libtiff-bad-write.tif  sqlite-bad-ptr3.sql  sqlite-null-ptr8.sql
bash-uninit-mem.var       libtiff-uninit-mem.tif  sqlite-heap-overflow.sql  sqlite-null-ptr9.sql
ffmpg-h264-bad-read.mp4  libtiff-uninit-mem2.tif  sqlite-heap-overflow.sql  sqlite-oob-read.sql
ffmpg-h264-bad-read.mp4  libtiff-uninit-mem3.tif  sqlite-negative-memset.sql  sqlite-oob-write.sql
ffmpg-h264-call-stack-overflow.mp4  libtiff-uninit-mem4.tif  sqlite-null-ptr1.sql  sqlite-stack-buf-overflow.sql
file-fpu-exception.elf    libxml2-bad-read.xml  sqlite-null-ptr10.sql  sqlite-stack-exhaustion.sql
firefox-bmp-leak.bmp      msie-dht-leak.jpg    sqlite-null-ptr11.sql  sqlite-unint-mem.sql
firefox-chrome-leak.jpg   msie-jxr-mem-leak.jpg  sqlite-null-ptr12.sql  sqlite-use-after-free.sql
firefox-gif-leak.gif      msie-png-mem-leak.png  sqlite-null-ptr13.sql  strings-bfd-badptr.elf
firefox-gif-leak2.gif     msie-tiff-mem-leak.tif  sqlite-null-ptr14.sql  strings-bfd-badptr2.elf
jxr-lib-crash.jxr          msie-elib-dos.png     sqlite-null-ptr15.sql  strings-stack-overflow
jxr-lib-crash2.jxr         openssl-null-ptr.der  sqlite-null-ptr2.sql   strings-unchecked-ctr.elf
jxr-lib-crash3.jxr         openssl-null-ptr2.der  sqlite-null-ptr3.sql   tcpdump-arp-crash.pcap
jxr-lib-crash4.jxr         photoshop-mem-leak.jpg  sqlite-null-ptr4.sql   tcpdump-ppp-crash.pcap
lesspipe-cpio-bad-write.cpio  sqlite-bad-free.sql    sqlite-null-ptr5.sql   unrtf-arbitrary-read.rtf
libjpeg-sos-leak.jpg       sqlite-bad-ptr.sql     sqlite-null-ptr6.sql   unzip-t-mem-corruption.zip
libjpeg-turbo-dht-leak.jpg  sqlite-bad-ptr2.sql    sqlite-null-ptr7.sql

./afl-2.57b/experimental:
README.experiments  asan_cgrouops  canvas_harness  crash_triage  libpng_no_checksum  post_library
argv_fuzzing        bash_shellshock  clang_asm_normalize  distributed_fuzzing  persistent_demo

./afl-2.57b/experimental/argv_fuzzing:
argv-fuzz-inl.h  test.c  test.txt
```

图1：ls 的基本、长格式、隐藏文件、人类可读大小、指定目录、按时间排序和递归查询。

```
id:000000,orig:seed id:000001,src:000000,op:havoc,rep:4,+cov
yaoliang@DESKTOP-HJNS067:~$ ls -R
.:
afl-2.57b  afl-exp  afl-test  test.txt

./afl-2.57b:
Android.bp          afl-as             afl-fuzz.c         afl-showmap        as                 libdislocator     types.h
CONTRIBUTING.md   afl-as.c          afl-g++           afl-showmap.c     config.h          libtkencaap
LICENSE            afl-as.h          afl-gcc           afl-tmin           debug.h           llvm_mode
Makefile            afl-clang          afl-gcc.c         afl-tmin.c         dictionaries      qemu_mode
README.md           afl-clang++       afl-gotcpu        afl-whatsup        docs              test-instr.c
afl-analyze        afl-cmin          afl-gotcpu.c      alloc-inl.h       experimental      test-libfuzzer-target.c
afl-analyze.c      afl-fuzz          afl-plot          android-ashmem.h   hash.h            testcases

./afl-2.57b/dictionaries:
README.dictionaries  html_tags.dict  js.dict  pdf.dict  png.dict  sql.dict  webp.dict
gif.dict             jpeg.dict      json.dict  perl.dict  regexp.dict  tiff.dict  xml.dict

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COPYING  QuickStartGuide.txt  life_pro_tips.txt  perf_tips.txt  technical_details.txt
Changelog  env_variables.txt  notes_for_asan.txt  sister_projects.txt  visualization
INSTALL  historical_notes.txt  parallel_fuzzing.txt  status_screen.txt  vuln_samples

./afl-2.57b/docs/visualization:
afl_gzip.png

./afl-2.57b/docs/vuln_samples:
bash-cmd-exec.var          libtiff-bad-write.tif  sqlite-bad-ptr3.sql  sqlite-null-ptr8.sql
bash-uninit-mem.var       libtiff-uninit-mem.tif  sqlite-heap-overflow.sql  sqlite-null-ptr9.sql
ffmpg-h264-bad-read.mp4  libtiff-uninit-mem2.tif  sqlite-heap-overflow.sql  sqlite-oob-read.sql
ffmpg-h264-bad-read.mp4  libtiff-uninit-mem3.tif  sqlite-negative-memset.sql  sqlite-oob-write.sql
ffmpg-h264-call-stack-overflow.mp4  libtiff-uninit-mem4.tif  sqlite-null-ptr1.sql  sqlite-stack-buf-overflow.sql
file-fpu-exception.elf    libxml2-bad-read.xml  sqlite-null-ptr10.sql  sqlite-stack-exhaustion.sql
firefox-bmp-leak.bmp      msie-dht-leak.jpg    sqlite-null-ptr11.sql  sqlite-unint-mem.sql
firefox-chrome-leak.jpg   msie-jxr-mem-leak.jpg  sqlite-null-ptr12.sql  sqlite-use-after-free.sql
firefox-gif-leak.gif      msie-png-mem-leak.png  sqlite-null-ptr13.sql  strings-bfd-badptr.elf
firefox-gif-leak2.gif     msie-tiff-mem-leak.tif  sqlite-null-ptr14.sql  strings-bfd-badptr2.elf
jxr-lib-crash.jxr          msie-elib-dos.png     sqlite-null-ptr15.sql  strings-stack-overflow
jxr-lib-crash2.jxr         openssl-null-ptr.der  sqlite-null-ptr2.sql   strings-unchecked-ctr.elf
jxr-lib-crash3.jxr         openssl-null-ptr2.der  sqlite-null-ptr3.sql   tcpdump-arp-crash.pcap
jxr-lib-crash4.jxr         photoshop-mem-leak.jpg  sqlite-null-ptr4.sql   tcpdump-ppp-crash.pcap
lesspipe-cpio-bad-write.cpio  sqlite-bad-free.sql    sqlite-null-ptr5.sql   unrtf-arbitrary-read.rtf
libjpeg-sos-leak.jpg       sqlite-bad-ptr.sql     sqlite-null-ptr6.sql   unzip-t-mem-corruption.zip
libjpeg-turbo-dht-leak.jpg  sqlite-bad-ptr2.sql    sqlite-null-ptr7.sql

./afl-2.57b/experimental:
README.experiments  asan_cgrouops  canvas_harness  crash_triage  libpng_no_checksum  post_library
argv_fuzzing        bash_shellshock  clang_asm_normalize  distributed_fuzzing  persistent_demo

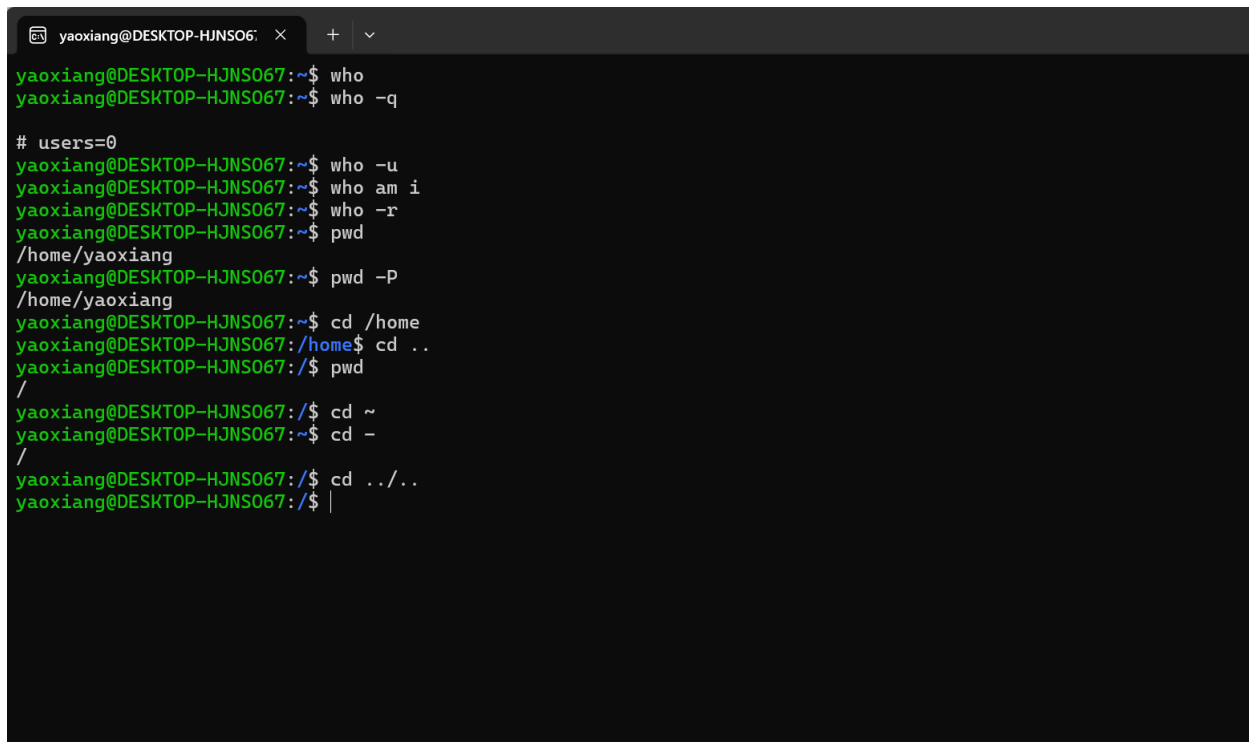
./afl-2.57b/experimental/argv_fuzzing:
argv-fuzz-inl.h  test.c  test.txt
```

图2：ls -R 的递归输出续页，说明该命令会遍历子目录。

2. who、pwd、cd：查看登录状态、当前目录与切换目录

实践了who、who -q、who -u、who am i、who -r、pwd、pwd -P、cd /home、cd ..、cd ~、cd - 和 cd ../..。who 系列大多没有显示用户并非命令失败：WSL

的当前终端会话通常没有写入传统的 utmp 登录记录，因此 `who -q` 显示 `# users=0`。`pwd` 和 `pwd -P` 均正确显示 `/home/yaoxiang`；各 `cd` 命令均成功完成目录切换。



```
yaoxiang@DESKTOP-HJNS067: ~$ who
yaoxiang@DESKTOP-HJNS067: ~$ who -q

# users=0
yaoxiang@DESKTOP-HJNS067: ~$ who -u
yaoxiang@DESKTOP-HJNS067: ~$ who am i
yaoxiang@DESKTOP-HJNS067: ~$ who -r
yaoxiang@DESKTOP-HJNS067: ~$ pwd
/home/yaoxiang
yaoxiang@DESKTOP-HJNS067: ~$ pwd -P
/home/yaoxiang
yaoxiang@DESKTOP-HJNS067: ~$ cd /home
yaoxiang@DESKTOP-HJNS067: /home$ cd ..
yaoxiang@DESKTOP-HJNS067: /$ pwd
/
yaoxiang@DESKTOP-HJNS067: /$ cd ~
yaoxiang@DESKTOP-HJNS067: ~$ cd -
/
yaoxiang@DESKTOP-HJNS067: /$ cd ../../
yaoxiang@DESKTOP-HJNS067: /$ |
```

图3：登录会话查询、路径显示和目录切换结果。

3. man：查看帮助手册

实践了 `man ls`、`man 5 passwd`、`man -k search` 和 `man man`。其中 `man -k search` 成功列出含 `search` 的手册条目。`man man` 报错 `can't resolve man7/groff_man.7`，原因是当前系统的 `man-db`/相关 `groff` 手册页面或索引不完整，导致该引用无法解析。这属于环境依赖缺失，可通过安装或修复相应手册包后再试。

```
yaoxiang@DESKTOP-HJNS067:~$ man ls
yaoxiang@DESKTOP-HJNS067:~$ man 5 passwd
yaoxiang@DESKTOP-HJNS067:~$ man -k search
apropos (1) - search the manual page names and descriptions
apt-patterns (7) - Syntax and semantics of apt search patterns
badblocks (8) - search a device for bad blocks
bsearch (3) - binary search of a sorted array
bzipgrep (1) - search possibly bzip2 compressed files for a regular expression
bzfgrep (1) - search possibly bzip2 compressed files for a regular expression
bzgrep (1) - search possibly bzip2 compressed files for a regular expression
find (1) - search for files in a directory hierarchy
git-bisect (1) - Use binary search to find the commit that introduced a bug
grep.c (1) - print PCRE patterns for searching C code
hsearch (3) - hash table management
hsearch_r (3) - hash table management
KEYCTL_SEARCH (2const) - search a keyring for a key
lfind (3) - linear search of an array
lsearch (3) - linear search of an array
lzgrep (1) - search possibly-compressed files for patterns
lzfgrep (1) - search possibly-compressed files for patterns
lzgrep (1) - search possibly-compressed files for patterns
manpath (1) - determine search path for manual pages
res_nsearch (3) - resolver routines
res_search (3) - resolver routines
strpbrk (3) - search a string for any of a set of bytes
tdelete (3) - manage a binary search tree
tdestroy (3) - manage a binary search tree
tfind (3) - manage a binary search tree
tsearch (3) - manage a binary search tree
twalk (3) - manage a binary search tree
twalk_r (3) - manage a binary search tree
wcschr (3) - search a wide character in a wide-character string
wcscspn (3) - search a wide-character string for any of a set of wide characters
wcsprbrk (3) - search a wide-character string for any of a set of wide characters
wcsrchr (3) - search a wide character in a wide-character string
wmemchr (3) - search a wide character in a wide-character array
xzegrep (1) - search possibly-compressed files for patterns
xzfgrep (1) - search possibly-compressed files for patterns
xzgrep (1) - search possibly-compressed files for patterns
zegrep (1) - search possibly compressed files for a regular expression
zfgrep (1) - search possibly compressed files for a regular expression
zgrep (1) - search possibly compressed files for a regular expression
yaoxiang@DESKTOP-HJNS067:~$ man man
man: can't resolve man7/groff_man.7
yaoxiang@DESKTOP-HJNS067:~$
```

图4：man -k search 成功；man man 因手册依赖缺失失败。

4. whereis、which：定位命令位置

whereis ls、whereis -b ls、whereis -m ls、which ls 和 which python3 均成功，分别找到了二进制程序和手册页路径。which cd 没有输出，不是异常：cd 是 Bash 内建命令，并非 PATH 中的独立可执行文件。应使用 type cd 或 help cd 查看其类型和帮助。

```
yaoxiang@DESKTOP-HJNS067:~$ whereis ls
ls: /usr/bin/ls /usr/share/man/man1/ls.1.gz
yaoxiang@DESKTOP-HJNS067:~$ whereis -b ls
ls: /usr/bin/ls
yaoxiang@DESKTOP-HJNS067:~$ whereis -m ls
ls: /usr/share/man/man1/ls.1.gz
yaoxiang@DESKTOP-HJNS067:~$ which ls
/usr/bin/ls
yaoxiang@DESKTOP-HJNS067:~$ which python3
/usr/bin/python3
yaoxiang@DESKTOP-HJNS067:~$ which cd
yaoxiang@DESKTOP-HJNS067:~$
```

图5：命令二进制文件、手册页和可执行路径查询。

5. find：按条件查找文件

实践了 `find /home -name "*.txt"`、`find . -type f`、`find . -type d`、`find . -size +10M` 和 `find . -mtime -2`。命令均成功执行；由于当前目录中包含 AFL 源码、构建目录和测试输出，递归查询的结果较多。`find . -size +10M` 找到大于 10MB 的归档和构建文件；`find . -mtime -2` 找到了最近两天修改的文件。

```
yaoxiang@DESKTOP-HJNSO67: ~$ find . -type d
./
./afl-exp/readelf-out/queue/id:001854,src:000121,op:havoc,rep:64
./afl-exp/readelf-out/queue/id:001896,src:000221,op:havoc,rep:128
./afl-exp/readelf-out/queue/id:000618,src:000000,op:flip2,pos:33308
./afl-exp/readelf-out/queue/id:000168,src:000000,op:flip1,pos:1033,+cov
./afl-exp/readelf-out/queue/id:000929,src:000000,op:arith8,pos:33521,val:+10
./afl-exp/readelf-out/queue/id:000597,src:000000,op:flip2,pos:3060
./afl-exp/readelf-out/queue/id:000714,src:000000,op:flip4,pos:33500
./afl-exp/readelf-out/queue/id:000770,src:000000,op:arith8,pos:18,val:-7,+cov
./afl-exp/readelf-out/queue/id:001399,src:000015,op:flip1,pos:3472,+cov
./afl-exp/readelf-out/queue/id:001402,src:000015,op:flip1,pos:3568,+cov
./afl-exp/readelf-out/queue/id:000004,src:000000,op:flip1,pos:5,+cov
./afl-exp/readelf-out/queue/id:000836,src:000000,op:arith8,pos:41,val:-32
./afl-exp/readelf-out/queue/id:001333,src:000012,op:flip1,pos:33520
./afl-exp/readelf-out/queue/id:001004,src:000000,op:int16,pos:17,val:+1024,+cov
./afl-exp/readelf-out/queue/id:001971,src:000285,op:ext_A0,pos:56
./afl-exp/readelf-out/queue/id:002095,src:000431,op:flip1,pos:32296
./afl-exp/readelf-out/plot_data
./afl-exp/readelf-out/fuzz_bitmap
./afl-exp/readelf-out/fuzzer_stats
./afl-exp/readelf-out/_cur_input
./landscape/sysinfo.log
./afl-test/test-instr
./afl-test/out/queue/id:000000,orig:seed
./afl-test/out/queue/id:000001,src:000000,op:havoc,rep:4,+cov
./afl-test/out/queue/.state/deterministic_done/id:000000,orig:seed
./afl-test/out/queue/.state/deterministic_done/id:000001,src:000000,op:havoc,rep:4,+cov
./afl-test/out/plot_data
./afl-test/out/fuzz_bitmap
./afl-test/out/fuzzer_stats
./afl-test/out/_cur_input
./afl-test/in/seed
./wget-hsts
yaoxiang@DESKTOP-HJNSO67:~$ find . -type d
./
./cache
./cache/ubuntu-pro
./cache/ubuntu-insights
./cache/ubuntu-insights/wsl_setup
./cache/ubuntu-insights/wsl_setup/local
./cache/ubuntu-insights/wsl_setup/uploaded
./cache/ubuntu-insights/linux
./cache/ubuntu-insights/linux/local
./cache/ubuntu-insights/linux/uploaded
./afl-2.57b
./afl-2.57b/.git
./afl-2.57b/.git/logs
./afl-2.57b/.git/logs/refs
./afl-2.57b/.git/logs/refs/remotes
./afl-2.57b/.git/logs/refs/remotes/origin
./afl-2.57b/.git/logs/refs/heads
./afl-2.57b/.git/refs
./afl-2.57b/.git/refs/remotes
```

图6：find 递归列出目录的输出过程。

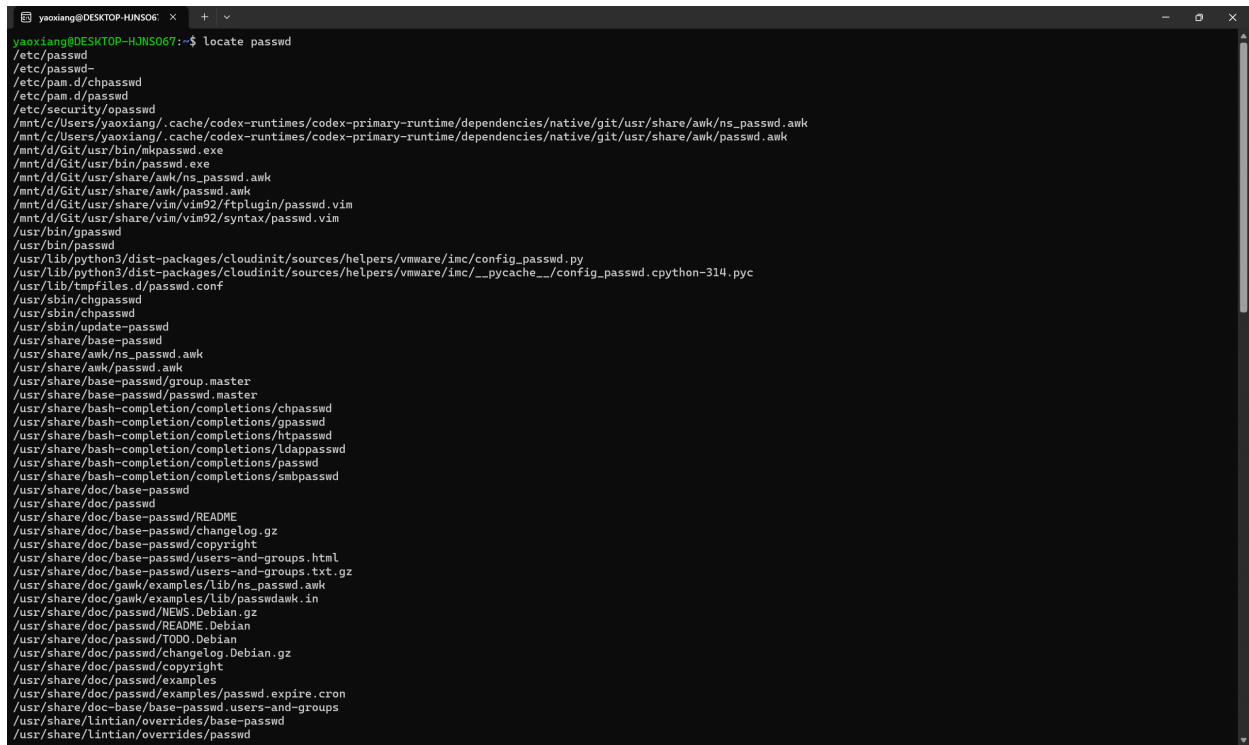
```
yaoxiang@DESKTOP-HJNSO67: ~$ find . -size +10M
./afl-exp/binutils-2.26-build/binutils/.deps
./afl-exp/binutils-2.26-build/binutils/doc
./afl-exp/binutils-2.26-build/ld
./afl-exp/binutils-2.26-build/ld/po
./afl-exp/binutils-2.26-build/ld/tmpdir
./afl-exp/binutils-2.26-build/ld/.libs
./afl-exp/binutils-2.26-build/ld/.deps
./afl-exp/binutils-2.26-build/ld/ldscripts
./afl-exp/binutils-2.26-build/bfd
./afl-exp/binutils-2.26-build/bfd/po
./afl-exp/binutils-2.26-build/bfd/.libs
./afl-exp/binutils-2.26-build/bfd/.deps
./afl-exp/binutils-2.26-build/bfd/doc
./afl-exp/readelf-out
./afl-exp/readelf-out/crashes
./afl-exp/readelf-out/queue
./afl-exp/readelf-out/queue/.state
./afl-exp/readelf-out/queue/.state/auto_extras
./afl-exp/readelf-out/queue/.state/variable_behavior
./afl-exp/readelf-out/queue/.state/deterministic_done
./afl-exp/readelf-out/queue/.state/redundant_edges
./afl-exp/readelf-out/hangs
./landscape
./afl-test
./afl-test/out
./afl-test/out/crashes
./afl-test/out/queue
./afl-test/out/queue/.state
./afl-test/out/queue/.state/auto_extras
./afl-test/out/queue/.state/variable_behavior
./afl-test/out/queue/.state/deterministic_done
./afl-test/out/queue/.state/redundant_edges
./afl-test/out/hangs
./afl-test/in
yaoxiang@DESKTOP-HJNSO67:~$ find . -mtime -2
./test.txt
./lessht
./cache/ubuntu-pro/ubuntu-pro.log
./motd_show
./bash_history
yaoxiang@DESKTOP-HJNSO67:~$
```

图7：按文件大小和修改时间查找的结果。

6. locate：基于数据库快速查找

实践了 `locate passwd`、`locate -c "*.conf"` 和 `locate -i`

`README`。命令均成功运行：`locate passwd` 返回了名称含 `passwd` 的路径，`locate -c "*.conf"` 统计为 398，`locate -i README` 在忽略大小写的条件下返回 `README` 相关路径。需要注意，`locate` 使用预先建立的数据库，新增文件可能需要运行 `updatedb` 后才会被查到。



```
yaoliang@DESKTOP-HJNS067:~$ locate passwd
/etc/passwd
/etc/passwd-
/etc/pam.d/chpasswd
/etc/pam.d/passwd
/etc/security/opasswd
/mnt/c/Users/yaoliang/.cache/codex-runtimes/codex-primary-runtime/dependencies/native/git/usr/share/awk/ns_passwd.awk
/mnt/c/Users/yaoliang/.cache/codex-runtimes/codex-primary-runtime/dependencies/native/git/usr/share/awk/passwd.awk
/mnt/d/Git/usr/bin/mkpasswd.exe
/mnt/d/Git/usr/bin/passwd.exe
/mnt/d/Git/usr/share/awk/ns_passwd.awk
/mnt/d/Git/usr/share/awk/passwd.awk
/mnt/d/Git/usr/share/vim/vim92/ftplugin/passwd.vim
/mnt/d/Git/usr/share/vim/vim92/syntax/passwd.vim
/usr/bin/gpasswd
/usr/bin/passwd
/usr/lib/python3/dist-packages/cloudinit/sources/helpers/vmware/imc/config_passwd.py
/usr/lib/python3/dist-packages/cloudinit/sources/helpers/vmware/imc/__pycache__/config_passwd.cpython-314.pyc
/usr/lib/tmpfiles.d/passwd.conf
/usr/sbin/chgpasswd
/usr/sbin/chpasswd
/usr/sbin/update-passwd
/usr/share/base-passwd
/usr/share/awk/ns_passwd.awk
/usr/share/awk/passwd.awk
/usr/share/base-passwd/group.master
/usr/share/base-passwd/passwd.master
/usr/share/bash-completion/completions/chpasswd
/usr/share/bash-completion/completions/gpasswd
/usr/share/bash-completion/completions/htpasswd
/usr/share/bash-completion/completions/ldappasswd
/usr/share/bash-completion/completions/passwd
/usr/share/bash-completion/completions/smbpasswd
/usr/share/doc/base-passwd
/usr/share/doc/passwd
/usr/share/doc/base-passwd/README
/usr/share/doc/base-passwd/changelog.gz
/usr/share/doc/base-passwd/copyright
/usr/share/doc/base-passwd/users-and-groups.html
/usr/share/doc/base-passwd/users-and-groups.txt.gz
/usr/share/doc/gawk/examples/lib/ns_passwd.awk
/usr/share/doc/gawk/examples/lib/passwdawk.in
/usr/share/doc/passwd/NEWS.Debian.gz
/usr/share/doc/passwd/README.Debian
/usr/share/doc/passwd/TODD.Debian
/usr/share/doc/passwd/changelog.Debian.gz
/usr/share/doc/passwd/copyright
/usr/share/doc/passwd/examples
/usr/share/doc/passwd/examples/passwd.expire.cron
/usr/share/doc-base/base-passwd.users-and-groups
/usr/share/lintian/overrides/base-passwd
/usr/share/lintian/overrides/passwd
```

图8：查找名称含 `passwd` 的文件。

```
yaoxiang@DESKTOP-HJNS067: ~$ locate -c "*.conf"
398
yaoxiang@DESKTOP-HJNS067:~$ locate -i README
/etc/X11/Xreset.d/README
/etc/alternatives/README
/etc/apparmor.d/local/README
/etc/apparmor.d/rsyslog.d/README
/etc/apparmor/report-ignore/README.denylist
/etc/chromy/conf.d/README
/etc/chromy/sources.d/README
/etc/cloud/cloud.cfg.d/README
/etc/fonts/conf.d/README
/etc/sudoers.d/README
/etc/sysctl.d/README.sysctl
/etc/terminfo/README
/home/yaoxiang/afl-2.57b/README.md
/home/yaoxiang/afl-2.57b/dictionaries/README.dictionaries
/home/yaoxiang/afl-2.57b/experimental/README.experiments
/home/yaoxiang/afl-2.57b/libdislocator/README.dislocator
/home/yaoxiang/afl-2.57b/libtokencap/README.tokencap
/home/yaoxiang/afl-2.57b/llvm_mode/README.llvm
/home/yaoxiang/afl-2.57b/qemu_mode/README.qemu
/home/yaoxiang/afl-2.57b/testcases/README.testcases
/home/yaoxiang/afl-exp/binutils-2.26/README
/home/yaoxiang/afl-exp/binutils-2.26/README-maintainer-mode
/home/yaoxiang/afl-exp/binutils-2.26/bfd/README
/home/yaoxiang/afl-exp/binutils-2.26/binutils/README
/home/yaoxiang/afl-exp/binutils-2.26/binutils/testsuite/binutils-all/windres/README
/home/yaoxiang/afl-exp/binutils-2.26/elfcpp/README
/home/yaoxiang/afl-exp/binutils-2.26/gas/README
/home/yaoxiang/afl-exp/binutils-2.26/gas/testsuite/gas/hppa/README
/home/yaoxiang/afl-exp/binutils-2.26/gas/testsuite/gas/i860/README.i860
/home/yaoxiang/afl-exp/binutils-2.26/gold/README
/home/yaoxiang/afl-exp/binutils-2.26/gprof/README
/home/yaoxiang/afl-exp/binutils-2.26/intl/README
/home/yaoxiang/afl-exp/binutils-2.26/ld/README
/home/yaoxiang/afl-exp/binutils-2.26/ld/emulparams/README
```

图9：统计 .conf 文件数量和忽略大小写查找 README。

7. grep：检索文本内容

实践了 `grep "root" /etc/passwd`、`grep -i "error" /var/log/syslog`、`grep -v "root" /etc/passwd`、`grep -r "nginx" /etc/` 和 `grep -n "bash" /etc/passwd`。除递归检索时的受限文件外，均成功运行。`grep -r "nginx" /etc/` 对部分敏感配置文件显示 `Permission denied`，原因是普通用户不能读取如 `shadow`、`sudoers` 等受保护文件；这说明递归检索已执行，但结果不完整。若业务确有授权，可使用 `sudo grep -r "nginx" /etc/`，或用 `2>/dev/null` 隐藏无权限文件的报错。

```
yaoxiang@DESKTOP-HJNSO6: ~$ grep "root" /etc/passwd
root:x:0:0:root:/root:/bin/bash
yaoxiang@DESKTOP-HJNSO6: ~$ grep -i "error" /var/log/syslog
2026-09-21T08:17:28.583707+08:00 DESKTOP-HJNSO67 ubuntu-insights[647]: 2026/09/21 08:17:28 WARN failed to run wsl.exe -v error="exit status 1"
2026-09-21T08:17:28.856241+08:00 DESKTOP-HJNSO67 ubuntu-insights[647]: 2026/09/21 08:17:28 ERROR failed to collect insights: insights write failed: report already exists for this
period
yaoxiang@DESKTOP-HJNSO6: ~$ grep -v "root" /etc/passwd
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-networkd:x:998:998:systemd Network Management:/:/usr/sbin/nologin
dhcpcd:x:996:996:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
messagebus:x:995:995:System Message Bus:/nonexistent:/usr/sbin/nologin
syslog:x:100:101::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:989:989:systemd Resolver:/:/usr/sbin/nologin
_chrony:x:988:988:Chrony Daemon:/var/lib/chrony:/usr/sbin/nologin
landscape:x:101:103:/:/var/lib/landscape:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin
yaoxiang:x:1000:1000::/home/yaoxiang:/bin/bash
yaoxiang@DESKTOP-HJNSO6: ~$ grep -r "nginx" /etc/
grep: /etc/security/opasswd: Permission denied
grep: /etc/gshadow: Permission denied
grep: /etc/gshadow: Permission denied
grep: /etc/sudoers.d/README: Permission denied
grep: /etc/landscape/client.conf: Permission denied
grep: /etc/chrony/chrony.keys: Permission denied
grep: /etc/credstore.encrypted: Permission denied
grep: /etc/shadow: Permission denied
grep: /etc/sudoers: Permission denied
grep: /etc/.pwd.lock: Permission denied
grep: /etc/ssl/private: Permission denied
grep: /etc/shadow: Permission denied
grep: /etc/credstore: Permission denied
grep: /etc/polkit-1/rules.d: Permission denied
yaoxiang@DESKTOP-HJNSO6: ~$ grep -n "bash" /etc/passwd
1:root:x:0:0:root:/root:/bin/bash
27:yaoxiang:x:1000:1000::/home/yaoxiang:/bin/bash
yaoxiang@DESKTOP-HJNSO6: ~$
```

图10：grep 的普通匹配、忽略大小写、反向匹配、递归匹配和显示行号。

8. echo：输出与重定向

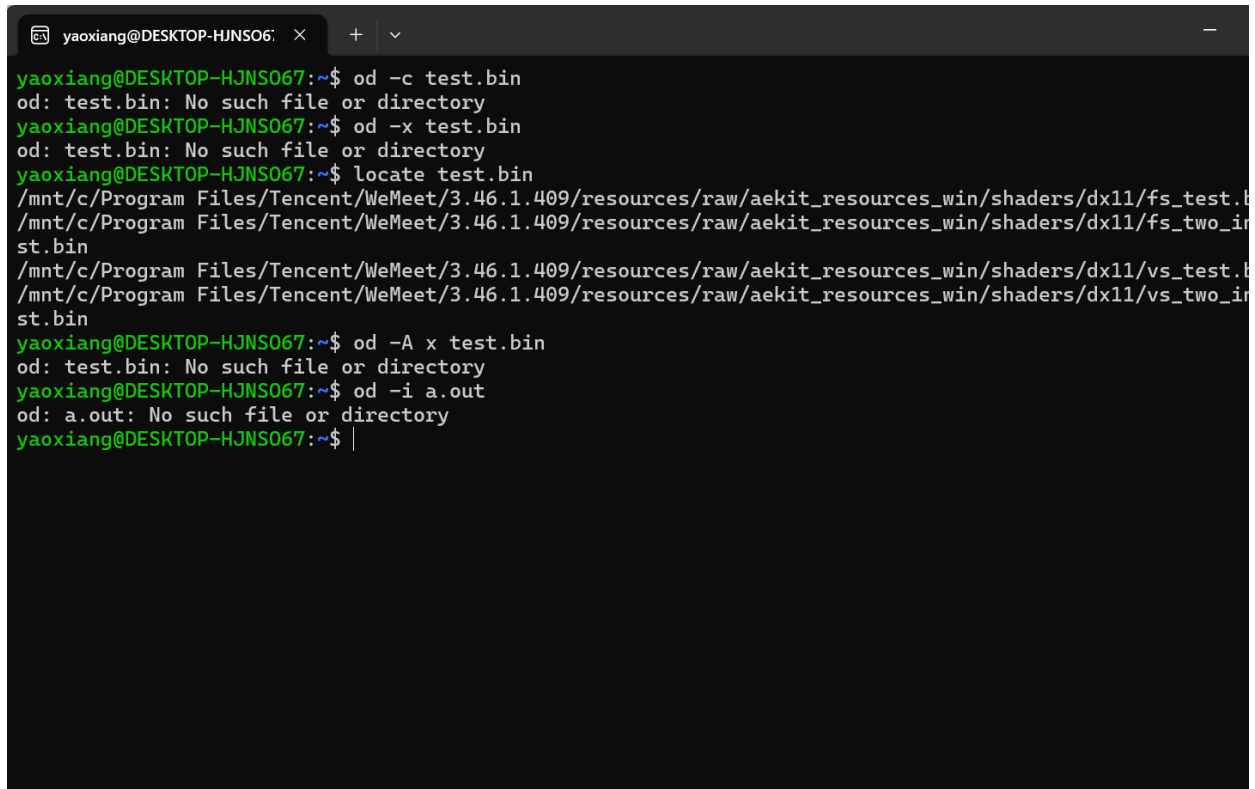
实践了 echo "Hello World"、echo \$HOME、echo "test content" > test.txt、echo -n "no newline" 和 echo *.md。前四项均成功；其中重定向成功创建并写入 test.txt，echo -n 按预期不输出行尾换行。echo *.md 显示字面量 *.md，不是命令失败，而是当前目录没有匹配的 Markdown 文件，Shell 未能展开通配符。

```
yaoxiang@DESKTOP-HJNSO6: ~$ echo "Hello World"
Hello World
yaoxiang@DESKTOP-HJNSO6: ~$ echo $HOME
/home/yaoxiang
yaoxiang@DESKTOP-HJNSO6: ~$ echo "test content" > test.txt
yaoxiang@DESKTOP-HJNSO6: ~$ cat test.txt
test content
yaoxiang@DESKTOP-HJNSO6: ~$ echo -n "no newline"
no newlineyaoxiang@DESKTOP-HJNSO6: ~$ echo *.md
*.md
yaoxiang@DESKTOP-HJNSO6: ~$
```

图11：echo 的普通输出、变量输出、重定向、取消换行和通配符示例。

9. od：以不同格式查看文件内容

实践了 `od -c test.bin`、`od -x test.bin`、`od -t d test.bin`、`od -A x test.bin` 和 `od -i a.out`。这些命令均因目标文件不存在而失败，提示 `No such file or directory`。locate `test.bin` 找到的是其他位置的同名文件，并不在当前工作目录，因此不能直接作为 `od` 的参数。后续可先创建测试文件，例如 `printf 'ABC' > test.bin`，再执行 `od -c test.bin`；`a.out` 则需要先编译 C 程序生成。

A terminal window with a dark background and light green text. The window title is 'yaoxiang@DESKTOP-HJNS067'. The terminal shows the following commands and their outputs:

```
yaoxiang@DESKTOP-HJNS067:~$ od -c test.bin
od: test.bin: No such file or directory
yaoxiang@DESKTOP-HJNS067:~$ od -x test.bin
od: test.bin: No such file or directory
yaoxiang@DESKTOP-HJNS067:~$ locate test.bin
/mnt/c/Program Files/Tencent/WeMeet/3.46.1.409/resources/raw/aekit_resources_win/shaders/dx11/fs_test.t
/mnt/c/Program Files/Tencent/WeMeet/3.46.1.409/resources/raw/aekit_resources_win/shaders/dx11/fs_two_in
st.bin
/mnt/c/Program Files/Tencent/WeMeet/3.46.1.409/resources/raw/aekit_resources_win/shaders/dx11/vs_test.t
/mnt/c/Program Files/Tencent/WeMeet/3.46.1.409/resources/raw/aekit_resources_win/shaders/dx11/vs_two_in
st.bin
yaoxiang@DESKTOP-HJNS067:~$ od -A x test.bin
od: test.bin: No such file or directory
yaoxiang@DESKTOP-HJNS067:~$ od -i a.out
od: a.out: No such file or directory
yaoxiang@DESKTOP-HJNS067:~$ |
```

图12：od 示例文件缺失导致的报错，以及 locate 的同名文件查询结果。

10. sort：文本排序

实践了 `sort names.txt`、`sort -n numbers.txt`、`sort -r names.txt`、`sort -k 2 -n data.txt` 和 `sort -u names.txt`。这些命令均因 `names.txt`、`numbers.txt`、`data.txt` 尚未在当前目录创建而失败，终端提示 `cannot read`。使用前应先准备对应的文本数据文件，例如用 `printf '■■■\n■■■\n■■■\n' > names.txt` 创建姓名文件。

```
yaoxiang@DESKTOP-HJNSO67:~$ od -c test.bin
od: test.bin: No such file or directory
yaoxiang@DESKTOP-HJNSO67:~$ od -x test.bin
od: test.bin: No such file or directory
yaoxiang@DESKTOP-HJNSO67:~$ locate test.bin
/mnt/c/Program Files/Tencent/WeMeet/3.46.1.409/resources/raw/aeKit_resources_win/shaders/dx11/fs_test.bin
/mnt/c/Program Files/Tencent/WeMeet/3.46.1.409/resources/raw/aeKit_resources_win/shaders/dx11/fs_two_input_test.bin
/mnt/c/Program Files/Tencent/WeMeet/3.46.1.409/resources/raw/aeKit_resources_win/shaders/dx11/vs_test.bin
/mnt/c/Program Files/Tencent/WeMeet/3.46.1.409/resources/raw/aeKit_resources_win/shaders/dx11/vs_two_input_test.bin
yaoxiang@DESKTOP-HJNSO67:~$ od -A x test.bin
od: test.bin: No such file or directory
yaoxiang@DESKTOP-HJNSO67:~$ od -i a.out
od: a.out: No such file or directory
yaoxiang@DESKTOP-HJNSO67:~$ sort names.txt
sort: cannot read: names.txt: No such file or directory
yaoxiang@DESKTOP-HJNSO67:~$ locate names.txt
/mnt/c/Users/yaoxiang/AppData/Local/Microsoft/Edge/User Data/ZxcvbnData/3.2.0.0/female_names.txt
/mnt/c/Users/yaoxiang/AppData/Local/Microsoft/Edge/User Data/ZxcvbnData/3.2.0.0/male_names.txt
/mnt/c/Users/yaoxiang/AppData/Local/Microsoft/Edge/User Data/ZxcvbnData/3.2.0.0/surnames.txt
/mnt/c/Users/yaoxiang/AppData/Local/Packages/MicrosoftWindows.Client.WebExperience_cw5n1h2txyewy/LocalState/EBWebView/ZxcvbnData/3.2.0.0/female_names.txt
/mnt/c/Users/yaoxiang/AppData/Local/Packages/MicrosoftWindows.Client.WebExperience_cw5n1h2txyewy/LocalState/EBWebView/ZxcvbnData/3.2.0.0/male_names.txt
/mnt/c/Users/yaoxiang/AppData/Local/Packages/MicrosoftWindows.Client.WebExperience_cw5n1h2txyewy/LocalState/EBWebView/ZxcvbnData/3.2.0.0/surnames.txt
/mnt/c/Users/yaoxiang/AppData/Local/Packages/OpenAI.Codex_2p2nqsd0c76g0/LocalCache/Roaming/Codex/web/Codex/ZxcvbnData/3/female_names.txt
/mnt/c/Users/yaoxiang/AppData/Local/Packages/OpenAI.Codex_2p2nqsd0c76g0/LocalCache/Roaming/Codex/web/Codex/ZxcvbnData/3/male_names.txt
/mnt/c/Users/yaoxiang/AppData/Local/Packages/OpenAI.Codex_2p2nqsd0c76g0/LocalCache/Roaming/Codex/web/Codex/ZxcvbnData/3/surnames.txt
yaoxiang@DESKTOP-HJNSO67:~$ sort -n numbers.txt
sort: cannot read: numbers.txt: No such file or directory
yaoxiang@DESKTOP-HJNSO67:~$ sort -r names.txt
sort: cannot read: names.txt: No such file or directory
yaoxiang@DESKTOP-HJNSO67:~$ sort -k 2 -n data.txt
sort: cannot read: data.txt: No such file or directory
yaoxiang@DESKTOP-HJNSO67:~$ sort -u names.txt
sort: cannot read: names.txt: No such file or directory
yaoxiang@DESKTOP-HJNSO67:~$
```

图13：sort 命令因测试数据文件缺失而无法读取。

三、问题记录与改进建议

1. 题目中“which cd 会显示对应信息”的表述不够准确。cd 是 Shell 内建命令，which 通常不会显示路径；使用 type cd 更合适。
2. od 和 sort 的示例依赖 test.bin、a.out、names.txt、numbers.txt、data.txt 等前置文件。练习说明应同时给出创建这些文件的步骤，避免把“文件不存在”误判为命令本身错误。
3. 在普通用户环境中，ls /root 与递归 grep /etc 出现权限不足是正常安全机制。应先理解权限边界，再在有明确授权时使用 sudo，不能为了消除报错而随意提升权限。
4. WSL 环境与完整 Linux 服务器存在差异，例如 who 可能显示 0 个登录用户，man 页面也可能因精简安装而缺少依赖。遇到这类情况应结合系统环境判断，不应只按预期输出判断成败。

四、学习感受

本次实践让我认识到 Linux 命令的学习不能只记住语法，还必须关注当前目录、文件是否存在、用户权限和系统环境。find、locate、grep 适合解决不同类型的检索问题：find 实时遍历目录，locate 查询数据库，grep 检索文件内容。后续我会在 openEuler 等国产操作系统环境中继续练习，并补齐 od、sort 所需的测试文件，以便观察正确的格式化查看和排序结果。